BML MUNJAL UNIVERSITY • SCHOOL OF ENGINEERING & TECHNOLOGY
Department of Computer Science & Engineering | AY 2026–27 (Batch 2023, 7th Semester)
Course: **CSE3101 — Agentic AI** (3 Credits, LDP: 2-0-2)

Course Faculty Mentors:
Dr. Soharab Hossain Shaikh • Mr. Pranshu Tiwari
Phase-2 Mid-Term Evaluation • October 2026

KAVACH (कवच): Mid-Term Comprehensive Capstone Report

Official Phase-2 Deliverables: Updated Problem Statement, Updated Plan, WIP Report, Code Guide & 1-Slider Coverage

Dhruv Jain (230532)
Lead Architect, FSM Orchestrator, MCP Server & Banking Highway

Hemant Goyal
Pre-Exec Guardrails, Entropy Engine, PyPI Slopsquatting Firewall

Rajat Yadav
Observability Telemetry, Groq Whisper Voice STT, Live QA & Demo

1. UPDATED PROBLEM STATEMENT (NEW CHANGES & THREAT EXPANSION)

In Phase-1, we defined four critical failure modes: supply-chain slopsquatting, credential/PII leakage, unbounded AST blast radius, and prompt injection loops. During Phase-2, as KAVACH evolved into an enterprise multi-agent highway, **four critical new threat vectors emerged**:

- **Financial Transaction Vulnerabilities (OWASP LLM01:2025):** Extending agents into core banking introduces transaction memo prompt injection, balance desynchronization without ACID atomicity, and violations of statutory RBI directives mandating High-Value Escrow quarantine for transfers exceeding ₹50,000.
- **Polyglot Supply-Chain Slopsquatting:** Hallucinated dependencies extend beyond Python to npm (`package.json`) and Go (`go.mod`), requiring a multi-ecosystem AST package firewall.
- **Statutory Indian DPDP Act 2023 Compliance:** Mandates zero-knowledge tokenization so that raw Aadhaar or PAN identifiers are never written even to internal database logs.
- **Cryptographic Tamper-Evidence:** Replaces modifiable SQLite/JSON audit trails with append-only NIST SHA3-512 Merkle DAGs for forensic non-repudiation.

2. UPDATED PROJECT PLAN & SLIP ANALYSIS (RUBRIC C6)

Milestone	Scope & Deliverables	Planned Week	Actual Week	Audit Status & Slips
Phase 1	Problem Definition, Charter, PEAS Matrix, Regex Scanners	Weeks 1–3	Weeks 1–3	**100% COMPLETED** (10/10 Score)
Vector RAG	Qdrant dense vector store + AST chunking	Weeks 4–5	Weeks 4–5	**100% COMPLETED** (test_rag.py PASS)
10-Stage FSM	Finite State Machine Orchestrator (orchestrator.py)	Week 6	Week 6	**100% COMPLETED** (10 Stages operational)
AST Blast Radius	Transitive reachability dependency closure B(v)	Week 7	Week 7	**100% COMPLETED** (< 12 ms latency)
PyPI Firewall	Slopsquatting interceptor with LRU cache	Week 8	Week 8	**100% COMPLETED** (HTTP 404 interceptor)
ReAct Sandbox	Bounded subprocess reflection loop ($N \leq 3$)	Week 9	Week 9	**100% COMPLETED** (5.0s watchdog timer)

Milestone	Scope & Deliverables	Planned Week	Actual Week	Audit Status & Slips
MCP & A2A	JSON-RPC 2.0 server & Agent2Agent message bus	Week 10	Week 10	**100% COMPLETED** (Cursor/Claude ready)
Banking Highway	Bharat Core Banking (/bank), SAGA & Time-Lock Escrow	Week 11	Week 11	**100% COMPLETED** (107/107 tests PASS)

DOCUMENTED SLIP MITIGATIONS (RUBRIC C6)

1. GPU VRAM Sizing: Running an unquantized 7B local Ollama model required >8GB VRAM causing OOMs on student laptops. We architected a dual-engine router (live Gemini 2.5 Flash + simulated air-gapped local fallback).
2. In-Repo Frameworks: Replaced bloated external pip packages (CrewAI, Google ADK) with native in-repo 8-hook callbacks and stateful Flows, cutting 1.8GB bloat and guaranteeing sub-millisecond execution.

3. WORK IN PROGRESS (WIP) REPORT: COURSE OUTCOME ALIGNMENT

Course Outcome	Curriculum Focus Statement	KAVACH Grounded Implementation	Verification Test Evidence
CO1	Foundational principles of intelligent agents and LLMs for multi-agent applications	• 10-Stage FSM Orchestrator (orchestrator.py) • In-repo Google ADK LlmAgent & CoordinatorAgent • Stateful Crew Flows with @start and @listen • Formal PEAS Modeling	test_kavach_100_suite.py (100/100 PASS) test_adk_triage.py (17/17 PASS) **100% VERIFIED**
CO2	Agentic design patterns and local LLMs for secure, privacy-preserving AI	• Shannon Entropy Scanner (H > 4.5) & Regex Filters • Zero-Knowledge DPDP Act Token Vault • Ephemeral Pytest Bounded ReAct Sandbox (N ≤ 3) • Air-Gapped Local Model Privacy Switch	test_security_v2.py (28/28 PASS) focused_check.py (5/5 PASS) Held-Out V2 F1 = 0.943 **100% VERIFIED**
CO3	Intelligent multi-agent, multimodal apps automating complex multi-step tasks	• Groq Whisper Multimodal Voice STT • Model Context Protocol (MCP) JSON-RPC 2.0 Server • AST Blast-Radius Reachability Graph B(v) • Bharat Core Banking Highway (/bank) SAGA Escrow	test_banking_100_suite.py (107/107 PASS) live_e2e_integration_test.py (11/11 PASS) test_mcp_protocol.py (8/8 PASS) **100% VERIFIED**

4. CODE & WORKING PROTOTYPE GUIDE (ONE-COMMAND VERIFICATION)

To verify the platform independently during viva and presentation evaluation:

- **Launch Unified Platform:** `.\run_kavach.ps1` or `python -m uvicorn 01_KAVACH_Platform.backend.app.main:app --host 127.0.0.1 --port 8000 --reload`
- **Master Live E2E Integration Suite (11/11 PASS in 4.0s):** `python verification/live_e2e_integration_test.py`
- **Focused Security Sandbox & Traversal Guards (5/5 PASS in 4.1s):** `python verification/focused_check.py`
- **KAVACH Deterministic 100 Suite (100/100 PASS in 0.38s):** `pytest 01_KAVACH_Platform/tests/test_kavach_100_suite.py -q`
- **Bharat Banking 100 Suite (107/107 PASS in 1.62s):** `pytest 02_Banking_and_Results_Dashboard/implementation_banks/test_banking_100_suite.py -q`
- **Mission Control Dashboard:** `http://localhost:8000` | **Bharat Banking Highway:** `http://localhost:8000/bank`

5. MASTER 1-SLIDER COVERAGE MATRIX SUMMARY (RUBRICS C1–C10)

FULL RUBRIC COMPLIANCE ATTESTATION (100% MET ACROSS ALL 10 CRITERIA)

C1 (Problem Definition): Formally expanded to banking prompt injections and DPDP Act compliance [MET].
C2 (Objectives & Outcomes): 142 endpoints, 0.017ms guardrail latency, and 300 test cases aligned to CO1, CO2, CO3 [MET].
C3 (Methodology): Multi-tier FSM pipeline with Qdrant vector retrieval, AST blast-radius, and slopsquatting firewall [MET].
C4 (Feasibility & Stack): Zero-cost student hardware execution with free Gemini API and offline deterministic fallback [MET].
C5 (Individual Understanding): Formal PEAS modeling and agent persona declarations across all team members

[MET].
C6 (Progress vs Plan): 100% Phase-2 milestones completed on schedule with documented slip mitigations [MET].
C7 (Technical Understanding): Shannon entropy $H > 4.5$, AST reachability $B(v)$, Verhoeff checksums [MET].
C8 (Problem Solving): Sub-millisecond deterministic filters (0.017 ms) chosen over slow LLM self-checks [MET].
C9 (Working Documentation): Single source of truth (METRICS.json), OpenAPI docs at /docs, and 86KB synopsis [MET].
C10 (Team Responsibility): Systematic role rotation across Phase-1, Phase-2, and Phase-3 complying with Handout Page 7 [MET].
